

Long Horizon: The Physical AI Summit, Oct 20-21, San Francisco. Join the waitlist.



What Is Named Entity Recognition? Selecting the Best Tool to Transform Your Model Training Data



Alexandre Bonnet

December 19, 2024 | 5 min read

Summarize with AI



What is Named Entity Recognition?

Named Entity Recognition (NER) is a fundamental task in **Natural Language Processing (NLP)** that involves locating and classifying named entities mentioned in unstructured text into predefined categories such as names, organizations, locations, dates, quantities, percentages, and monetary values. NER serves as a foundational component in various NLP applications, including information extraction, question answering, machine translation, and sentiment analysis.

At its core, NER processes textual data to identify and categorize key information. For example, in the sentence

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

Geopolitical entity (GPE), and "\$1 billion" as a Monetary value (MONEY).

Apple **ORG** is looking at buying U.K. **GPE** startup for \$1 billion **MONEY** .

Named Entity Recognition (NER) Example

How NER Works

The NER process identifies and classifies key information (entities) in text into predefined categories such as names, organizations, locations, dates, and more. The following are the general steps of the NER process:

Step #1: Text Input

The process begins with raw text data that needs to be analyzed.

"Apple Inc. is planning to open a new office in San Francisco in March 2025."

Step #2: Text Preprocessing

This step involves preparing the text for analysis by performing following operations.

Tokenization

Splitting the text into individual units called tokens (words, punctuation, etc.).

["Apple", "Inc.", "is", "planning", "to", "open", "a", "new", "office", "in", "San", "Francisco", "in", "March", "2025", "."]

Part-of-Speech Tagging

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

"VB"), ("a", "DT"), ("new", "JJ"), ("office", "NN"), ("in", "IN"), ("San", "NNP"), ("Francisco", "NNP"), ("in", "IN"), ("March", "NNP"), ("2025", "CD"), (".", ".")]

Step #3: Feature Extraction

Deriving relevant features from the tokens to assist the NER model in making accurate predictions.

- **Contextual Features:** Considering surrounding words to understand the context.
- **Orthographic Features:** Examining capitalization, punctuation, and numerical patterns.
- **Lexical Features:** Utilizing dictionaries or gazetteers to match known entity names.

Step #4: Model Application

Applying a trained NER model to classify each token (or group of tokens) into predefined entity categories.

- **Machine Learning Models:** Using algorithms like Conditional Random Fields (CRFs) or neural networks trained on annotated datasets.
- **Rule-Based Systems:** Employing handcrafted rules and patterns for specific entity types.

Step #5: Entity Classification

Assigning labels to tokens based on the model's predictions.

[("Apple Inc.", "ORG"), ("San Francisco", "LOC"), ("March 2025", "DATE")]

Step #6: Post-Processing

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

"Jordan" could refer to a person's name or a country; context is used to decide the correct classification.

Or, identified nested entities (entities within entities), such as a person's name within an organization. For example

"President [Barack Obama] of [the United States]"

Step #7: Output Generation

Producing the final annotated text with entities highlighted or in a structured format like JSON or XML.

Labels and Tagging Schemes in NER

Labels in NER

In NER, labels are the categories assigned to words or phrases identified as named entities within a piece of text. These labels indicate the type of entity detected, such as a person, organization, location, or date. The labeling process allows unstructured text to be converted into structured data, which can be used for various applications like information retrieval, question answering, and data analysis.

The set of labels used in NER can vary depending on the specific application, domain, or dataset. However, some standard labels are widely used across different NER systems:

Labels	Description	Example
Person (PER)	Names of people or fictional characters.	Albert Einstein," "Marie Curie," "Sherlock

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

Location (LOC)	Names of geographical places such as cities, countries, mountains, rivers.	"Mount Everest," "Nile River," "Paris."
Geo-Political Entity (GPE)	Geographical regions that are also political entities.	"United States," "Germany," "Tokyo."
Date	Expressions of calendar dates or periods.	"January 1, 2022," "the 19th century," "2010–2015."
Time	Specific times within a day or durations.	"5 PM," "midnight," "two hours."
Money	Monetary values, often accompanied by currency symbols.	"\$100," "€50 million," "1,000 yen."
Percent	Percentage expressions.	"50%," "3.14%," "half."
Facility (FAC)	Buildings or infrastructure.	"Eiffel Tower," "JFK Airport," "Golden Gate Bridge."
Product	Objects, vehicles, software, or any tangible items.	"iPhone," "Boeing 747," "Windows 10."
Event	Named occurrences such as wars, sports events, disasters.	"World War II," "Olympics," "Hurricane Katrina."
Work of Art	Titles of books, songs, paintings, movies.	"Mona Lisa," "To Kill a Mockingbird," "Star Wars."
Language	Names of languages.	"English," "Mandarin," "Spanish."

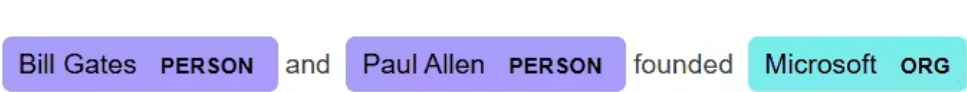
Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

PERSON (Nationality, Religious, or Political Group)	Nationalities, religious groups, or political affiliations.	"American," "Christians," "Democrat."
---	---	---------------------------------------

For example, in the following sentence:

Bill Gates and Paul Allen founded Microsoft

Bill Gates and Paul Allen recognized and classified as a PERSON entity and Microsoft is classified as an ORG (organization).



Tagging Schemes in NER

In addition to the entity labels, NER systems often use tagging schemes to indicate the position of words within entities. The most common schemes are:

BIO Tagging (Begin, Inside, Outside)

Tags	Description
B-XXX	Beginning of an entity of type XXX
I-XXX	Inside (continuation) of an entity of type XXX
O	Outside any named entity

Example

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

IOBES Tagging (Inside, Outside, Begin, End, Single)

Tags	Description
B-XXX	Beginning of an entity of type XXX
I-XXX	Inside (continuation) of an entity of type XXX
E-XXX	End of an entity of type XXX
S-XXX	Single-token entity
O	Outside any named entity

Example



IOB2

This tagging is similar to BIO but it ensures that the beginning of every entity is marked with a B- tag, even if it immediately follows another entity of the same type.

Example:

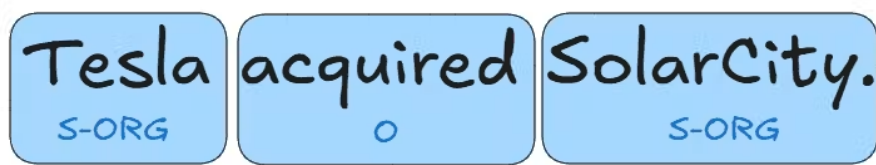


In this case, "Apple" is tagged as the beginning of an organization (B-ORG), and "U.K." is tagged as the beginning of a location (B-LOC).

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

Tags	Description
B- (Beginning)	First token of a multi-token entity.
I- (Inside)	Tokens inside a multi-token entity.
E- (End)	Last token of a multi-token entity.
S- (Single)	Single-token entity.
O (Outside)	Tokens not part of any entity.

Example:



Here, both "Tesla" and "SolarCity" are single-token entities tagged as S-ORG.

Domain-Specific Labels

In specialized domains, additional labels may be used to capture domain-specific entities. For example in the biomedical domain, the labels such as Gene/Protein, Disease, Chemical, Drug are used.



Similarly in financial domain labels such as Financial Instrument, Market Index, Economic Indicator etc. are used.



Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

Rule-Based Methods

Rule-based NER systems rely on manually specified linguistic rules and patterns to identify entities. These rules often utilize regular expressions, dictionaries (gazetteers), and part-of-speech tagging to detect predefined entity types. For example, a rule might specify that a capitalized word followed by "Inc." or "Ltd." should be classified as an organization. While rule-based methods can achieve high precision in specific domains, they often suffer from limited recall and are not easily scalable to diverse or evolving datasets. Additionally, developing and maintaining these rules can be labor-intensive and may not generalize well to new or informal text sources.

Machine Learning-Based Methods

Machine learning approaches involve training statistical models on annotated datasets to automatically recognize entities. Algorithms such as Conditional Random Fields (CRFs) and Support Vector Machines (SVMs) have been commonly used in this context. These models learn to identify entities based on features extracted from the text, such as word shapes, context words, and syntactic information. Machine learning methods generally offer better adaptability to different domains compared to rule-based systems and can handle a wider variety of entity types. However, they require substantial amounts of labeled training data and may still struggle with recognizing entities in noisy or informal text.

Deep Learning-Based Methods

Deep learning based methods use neural networks to capture complex patterns in data. Models such as **Recurrent Neural Networks (RNNs)**, **Long Short-Term Memory (LSTM) networks**, and **Transformers** (e.g., **BERT**) have been used to understand the text. These models can automatically learn feature representations from raw text, reducing the need for manual feature

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

Hybrid Approaches

Hybrid NER systems combine elements of rule-based, machine learning, and deep learning methods to use the advantages of each. For example, a hybrid system might use rule-based techniques to preprocess text and identify obvious entities, followed by a machine learning model to detect more complex cases. Alternatively, deep learning models can be supplemented with domain-specific rules to improve accuracy in specialized fields. Hybrid approaches aim to balance precision and recall while maintaining flexibility across different domains and text types.

Each of these approaches has its own set of trade-offs concerning accuracy, scalability, and resource requirements. The choice of method often depends on the specific application, the availability of labeled data, and the computational resources at hand.

Evaluation Metrics for NER

Evaluating a NER model is essential to measure its ability to accurately identify and classify entities. The evaluation metrics typically focus on Precision, Recall, and F1-Score, which are calculated based on the comparison between the predicted entities and the actual entities in the dataset.

Precision

Precision measures the proportion of entities predicted by the model that are correct. High precision indicates that the model makes fewer false positive errors.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

or the relevant entities.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1-Score

The F1-Score is the harmonic mean of Precision and Recall, providing a single score that balances the two. High F1-Score suggests a good balance between precision and recall.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Evaluating an NER Model

Consider the following example:

Apple Inc. is planning to open a new office in San Francisco in March 2025.

Ground Truth (Actual Entities):

Apple Inc.	ORGANIZATION
San Francisco	LOCATION
March 2025	DATE

Model Prediction:

Apple Inc.	ORGANIZATION  (True Positive)
San Francisco	LOCATION  (True Positive)

Long Horizon: The Physical AI Summit, Oct 20-21, San Francisco. Join the waitlist.

Calculation:

True Positives (TP)	2 (Apple Inc., San Francisco)
False Positives (FP)	1 (office)
False Negatives (FN)	1 (March 2025)

Metrics:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 2 / (2 + 1) = 0.67$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 2 / (2 + 1) = 0.67$$

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall} / \text{Precision} + \text{Recall}) = 2 \times (0.67 \times 0.67 / 0.67 + 0.67) = 0.67$$

Tools for Transform data for NER

Transforming data for NER involves converting raw text into a structured, annotated format suitable for model training. Various tools are available for this task, each offering unique features to facilitate the process. Below is a detailed explanation of tools that help transform data for NER:

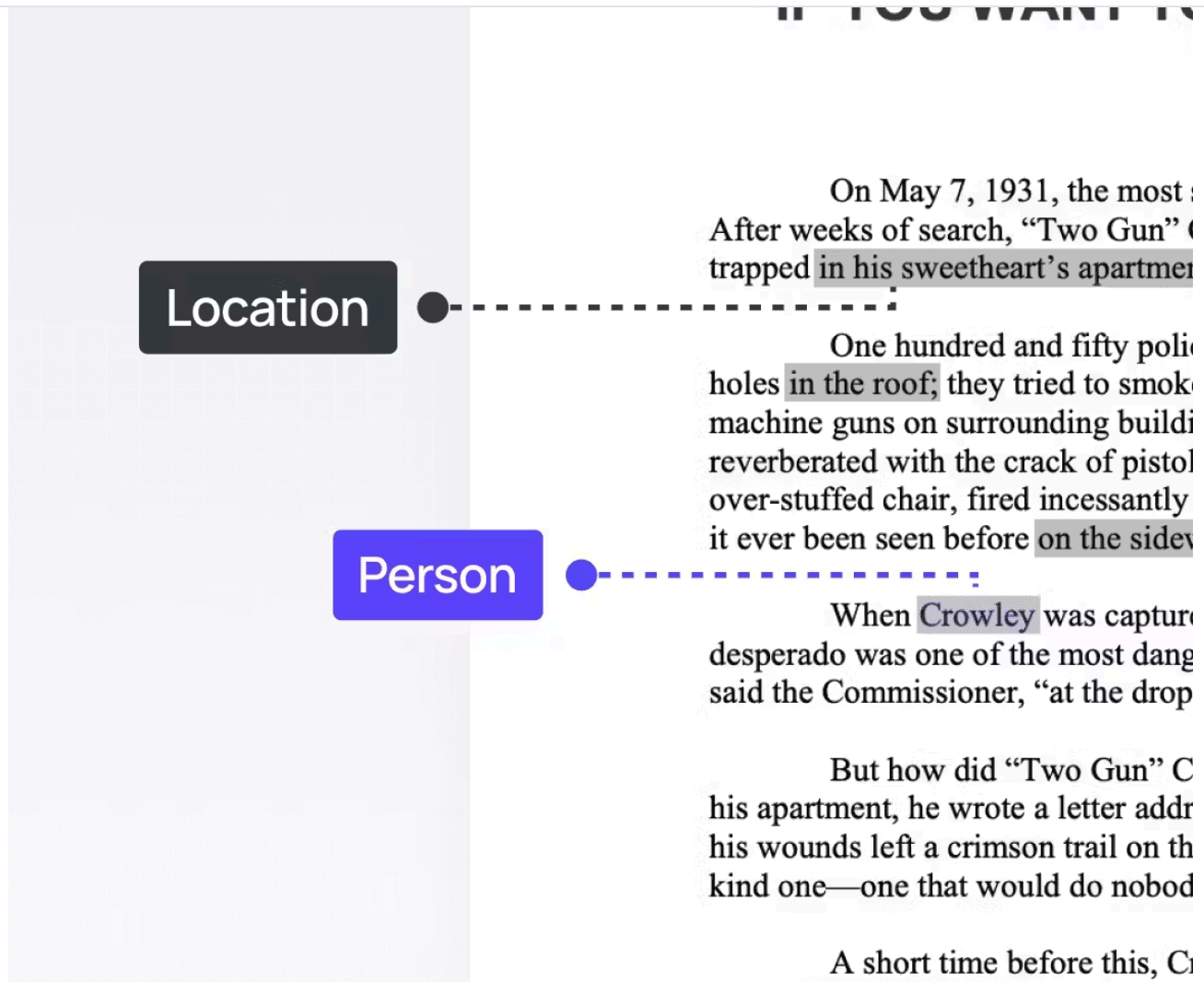
Encord

Encord is an AI data development platform for managing, curating and annotating large-scale text and document datasets, as well as evaluating LLM performance. AI teams can use Encord to label document and text files containing text and complex images and assess annotation quality using several metrics. The platform has robust cross-collaboration functionality across:

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

- **Encord Annotate:** Leverage SOTA AI-assisted labeling workflows and flexibly setup complex ontologies to efficiently and accurately label largescale document and text datasets for training, fine-tuning and aligning AI models at scale.
- **Encord Active:** Evaluate and validate AI models to surface, curate, and prioritize the most valuable data for training and fine-tuning to supercharge AI model performance. Leverage automatic reporting on metrics like mAP, mAR, and F1 Score. Combine model predictions, vector embeddings, visual quality metrics and more to automatically reveal errors in labels and data.

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.



NER annotation in Encord ([Source](#))

Doccano

Doccano is an open-source, user-friendly annotation tool for text labeling tasks which also supports NER annotation. It has following features:

- Intuitive interface for labeling text spans.
- Support for sequence labeling (NER), text classification, and translation tasks.
- Collaborative annotation for teams.

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

Donald John Trump (born June 14, 1946) is the 45th and current president of the United States. Before entering politics, he was a businessman and television personality. Trump was born and raised in the New York City borough of Queens, and received a B.S. degree in economics from the Wharton School at the University of Pennsylvania. He took charge of his family's real-estate business in 1971, renamed it The Trump Organization, and expanded its operations from Queens and Brooklyn into Manhattan. The company built or renovated skyscrapers, hotels, casinos, and golf courses. Trump later started various side ventures, mostly by licensing his name. He owned the Miss Universe and Miss USA beauty pageants from 1996 to 2015, and produced and hosted The Apprentice, a reality television show, from 2003 to 2015. Forbes estimates his net worth to be \$3.1 billion.

Key	Value
wikiPageID	4848272
Born	1946
Political party	Republican
Spouse	Melania Knauss
Parents	Fred Trump, Mary Anne MacLeod
Residence	White House

Named Entity Recognition

Prodigy

Prodigy is a commercial, Python-based annotation tool designed for machine learning workflows and can be used for NER annotations. It has following features:

- Active learning to prioritize uncertain samples for annotation.
- Seamless integration with spaCy models.
- Support for manual annotation, model-in-the-loop annotation, and rule-based labeling.
- Flexible export formats for training data

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

VIEW IDner_manual

PROGRESS

THIS SESSION0

TOTAL0

0%

ACCEPT0

REJECT0

IGNORE0

HISTORY

© 2017-2024 Explosion (Prodigy v1.11)

infected mosquitoes and give them small doses of radiation. She explains that the levels have to be exact because if the mosquitoes receive too much radiation, the results could be more devastating than the virus itself.

SOURCE: CMU Movie Summary Corpus

✓

✗

⊘

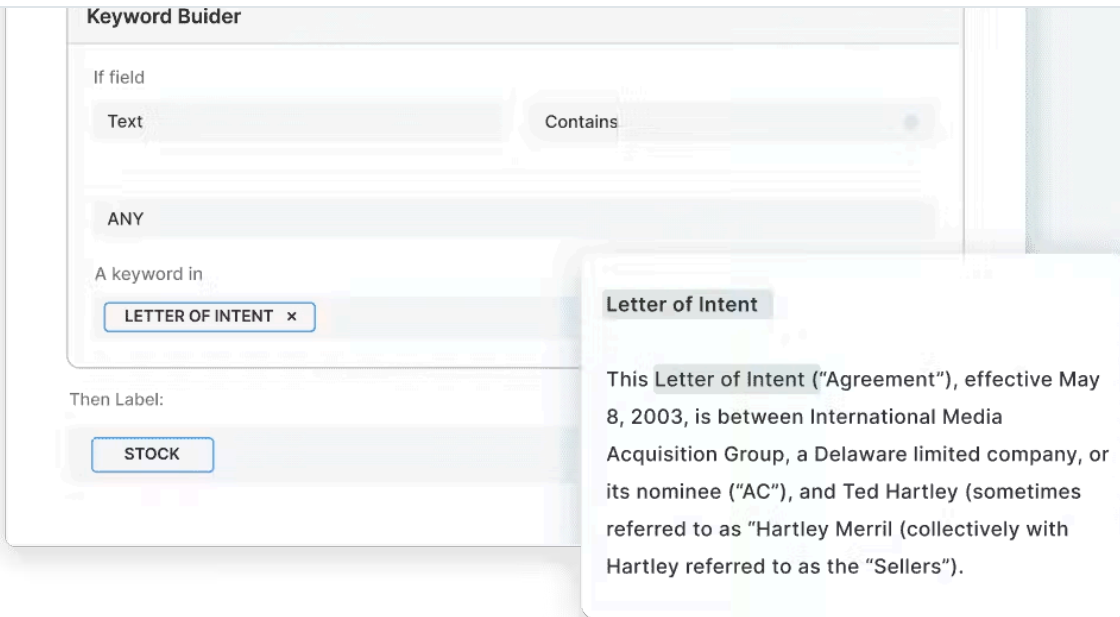
↩

Snorkel

Snorkel is a data programming platform for programmatically labeling and transforming training data. It supports many annotation tasks including support for NER annotation. It has following features:

- Create labeling functions to annotate data programmatically.
- Combines weak supervision signals to generate probabilistic labels.
- Scalable and suitable for large datasets.

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.



Snorkel NER annotation ([Source](#))

spaCy

spaCy is a popular NLP library in Python. It also provides options for training and evaluating NER models. It has following features:

- Pre-trained models for entity recognition.
- Supports custom NER annotation and training pipelines.
- Integration with Prodigy for annotation tasks.

The Mars Orbiter Mission **PRODUCT** (**MOM** **ORG**), informally known as **Mangalyaan** **GPE** , was launched into **Earth** **LOC** orbit on **5 November 2013** **DATE** by **the Indian Space Research Organisation** **ORG** (**ISRO** **ORG**) and has entered **Mars** **LOC** orbit on **24 September 2014** **DATE** . **India** **GPE** thus became the **first** **ORDINAL** country to enter **Mars** **LOC** orbit on its **first** **ORDINAL** attempt. It was completed at a record low cost of **\$74 million** **MONEY** .

spaCy NER example ([Source](#))

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

- Pre-trained models for NER in multiple languages.
- Tools for training custom NER models using labeled data.
- Support for tokenization, sentence segmentation, and other preprocessing tasks.

Luke Rawlence PERSON joined **Aiimi** ORG as a data scientist in **Milton Keynes** PLACE, after finishing his computer science degree at the **University of Lincoln**. ORG

NER in OpenNLP ([Source](#))

Stanza

Stanza is a Python NLP library developed by Stanford NLP Group. It supports multilingual NER and provides different **NER models**. It has following features:

- Pre-trained NER models for multiple languages.
- Easy integration with Python workflows.

```
import stanza

nlp = stanza.Pipeline(lang='en', processors='tokenize,ner')
doc = nlp("Chris Manning teaches at Stanford University. He lives in the Bay Area.")
print(*[f'entity: {ent.text}\ttype: {ent.type}' for sent in doc.sentences for ent in sent.ents], sep='\n')
```

```
entity: Chris Manning    type: PERSON
entity: Stanford University    type: ORG
entity: the Bay Area     type: LOC
```

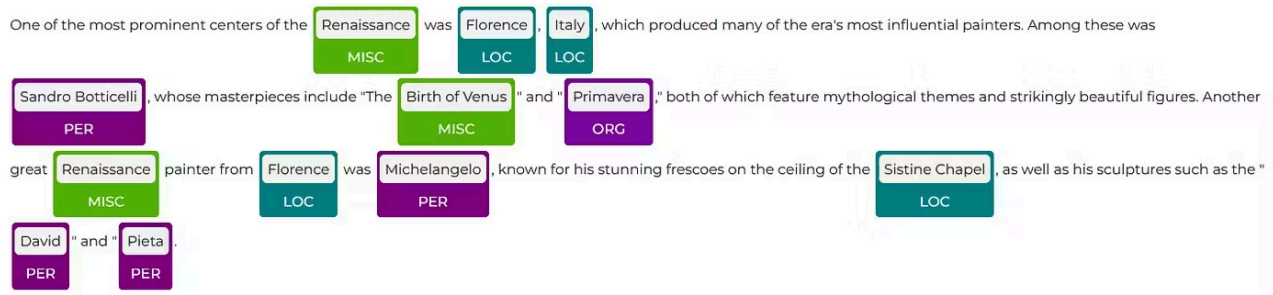
Stanza NER example ([Source](#))

Spark NLP

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

We trained NER models for large-scale text processing.

- Supports training custom models for NER tasks.
- Integration with other Spark-based tools.



Spark NLP example ([Source](#))

How Encord helps in NER data annotation

Encord supports various data types, including text, making it suitable for NER annotation tasks. It helps in managing, annotating, and iterating on training data for machine learning tasks. Here is how Encord helps in the NER annotation:

Intuitive Annotation Interface

Encord offers a user-friendly text annotation interface, making it easy for annotators to highlight and label text spans as entities. It helps in highlighting text directly to label it as an entity. Annotators can highlight specific words or phrases within the text. Annotators can assign entity labels, such as PERSON, LOCATION, ORGANIZATION, DATE, or any other custom tag defined in the ontology.

Ontology Management

Encord allows you to define a clear and structured **ontology** for your NER project. This ontology ensures consistent labeling and defines the entity types and their attributes. Users can create custom ontologies for specific projects

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

Encord supports team-based annotation projects. It allows multiple annotators to work on the same dataset while maintaining consistency. It enables project managers or reviewers to check and approve annotations using built-in review **workflows**. It supports multi-stage review processes to help ensure high-quality labels.

Model-Assisted Annotation

Encord integrates with **pre-trained models** or custom machine learning (ML) models to assist annotators by providing pre-annotations. Annotators can validate, correct, or refine these predictions, significantly reducing manual workload. In Encord you can import a pre-trained NER model (e.g., spaCy, Hugging Face Transformers) and use the model to generate initial predictions on raw text. Annotators review and validate these suggestions, correcting any inaccuracies.

Multi-Modality Support

Encord platform supports annotation of different types of data including images, videos, and **multi-modal** datasets. This is particularly useful for cross-domain projects where text is tied to visual data. For example, in medical applications annotating entities like SYMPTOM and DIAGNOSIS in patient text reports alongside CT scans or X-rays. Similarly in multimedia data, extracting named entities from speech transcriptions in videos and linking them to visual metadata can be easily done in Encord.

Export and Integration

Encord makes it easy to export annotated data in formats compatible with popular NLP frameworks and tools such as spaCy, Hugging Face Transformers, TensorFlow and many more. The supported formats are JSON, CSV, JSONL

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

NER identifies entities such as names, organizations, locations, and more within unstructured text accurately, but it may also face challenges. Following are some of the challenges in NER.

Ambiguity

Ambiguity arises when a word or phrase can have multiple meanings depending on its context. NER models can struggle to correctly classify such entities, especially in the absence of sufficient context. There are two main types of ambiguity:

- **Lexical Ambiguity:** Words that can belong to multiple categories (e.g., person, organization, or location).
- **Contextual Ambiguity:** Entities that require surrounding text to determine their exact type.

Example:

Sentence: "I visited Jordan last summer to attend the Jordan Shoes event."

Jordan (First occurrence): Refers to a location (country).

Jordan Shoes: Refers to an organization (brand name).

Context-sensitive words require language models capable of understanding relationships in the text. Traditional rule-based models struggle with ambiguous entities due to limited contextual awareness.

Nested Entities

Nested entities occur when one entity is embedded within another, creating hierarchical structures. This challenge is common in domains like legal,

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

Sentence: "The University of California, Berkeley is a top-ranked university."

University of California: Organization (outer entity).

Berkeley: Location (nested entity within the organization name).

Traditional NER models often assume that entities do not overlap, leading to errors when an entity is nested. Nested structures require advanced models that can handle multiple layers of entities (e.g., transformer-based approaches or dependency parsers).

Entity Boundary Detection

Entity boundary detection involves identifying the exact start and end positions of an entity. Errors can occur when entities contain compound phrases or when boundaries are unclear.

Example:

Sentence: "New York City Mayor Eric Adams introduced a new policy."

Correct Entity: "Eric Adams" → (PERSON)

Incorrect Boundary: "New York City Mayor Eric" → (Partial extraction)

Compound entities or multi-word entities can confuse models. Entity boundaries may vary depending on language structure and dataset consistency.

Domain-Specific Entities

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

Example:

Sentence: "The patient was prescribed metformin for controlling Type 2 diabetes."

Entities: "metformin" -> (MEDICATION), "Type 2 diabetes" -> (DIAGNOSIS)

General-purpose models may not recognize "metformin" or "Type 2 diabetes" as entities.

Entities in specialized domains require custom tagging schemas and training data. Annotating large domain-specific datasets is time-consuming and expensive.

Language and Morphological Variations

NER models may face challenges with languages that have complex grammatical structures, lack capitalization cues, or feature multiple inflected forms of words.

Example:

Capitalization Issues (Lowercase or noisy text):

Sentence: "steve jobs was the co-founder of apple inc."

Challenge: Models relying on capitalization may miss "steve jobs" as a PERSON.

Some languages (e.g., German, Finnish) have inflected words, where entity names can change forms depending on usage. Standard NER models trained on English datasets may struggle with non-English text without additional training.

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

- The NER process involves text preprocessing, feature extraction, and contextual analysis using models.
- NER uses tagging schemes like BIO (Begin–Inside–Outside) to mark entity boundaries.
- NER tools help annotate training data for models. Popular tools include Encord, Prodigy, and Doccano.
- NER is used in information extraction, chatbots, customer feedback analysis, and healthcare and in many other applications.
- Tools like Encord simplify annotation, making it easier to build accurate NER models.

If you're extracting images and text from PDFs to build a dataset for your multimodal AI model, be sure to explore [Encord's Document Annotation Tool](#)—to train and fine-tune high-performing NLP Models and LLMs.

< PREVIOUS

How to Implement Audio File
Classification: Categorize an...

NEXT >

Exploring Google DeepMind's
Latest AI Innovations: Gemini

Frequently asked questions

What is Named Entity Recognition (NER)?



Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

What are common applications of NER?



Why is NER important for NLP?



How does Encord handle automated annotations and what are its benefits?



What are the capabilities of Encord's annotation platform compared to open source solutions?



How can Encord help future-proof annotation processes compared to in-house solutions?



What features does Encord have to speed up the annotation process?



What is the process for extracting meaningful information from unstructured data using Encord?



What types of text-based annotations can be performed using Encord?



What types of annotation tools can be utilized within Encord?



Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. [Join the waitlist.](#)

What features does Encord offer to enhance annotation efficiency?



Is it possible to gradually reduce manual annotation efforts using Encord's platform?



Get the data right.

300+ of the best AI teams in the world use Encord.

[Take a tour](#)

[Book a demo](#)



[Terms](#) • [Privacy Policy](#)

Subscribe to our newsletter

Get the latest research and news for ML leaders in
your inbox.

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

Platform

PRODUCT

Annotation and labeling

Post-training alignment

Data indexing & curation

Data agents

INTEGRATIONS

Merlin

Cloud integrations

Security

SERVICES

Annotation services

Data collection services

Physical AI data services

Solutions

PHYSICAL AI

Autonomous Vehicles & ADAS

Robotics & Humanoids

Drones & Aerial Autonomy

Industrial & Manufacturing

ENTERPRISE

Surgical video

ID verification

Smart cities & security

Sports AI

Defence

Voice AI

Healthcare

Frontier & generative AI

MODALITIES

Multimodal

Image

Video

Long Horizon: The Physical AI Summit, Oct 20–21, San Francisco. Join the waitlist.

DICOM & NIfTI

Resources

COMPANY

Pricing

Customers

About us

Careers

Press

Contact us

ENGAGE

In-person events

Webinars

Glossary

Blog

Documentation

E-MM1 dataset



© 2026 Encord. All rights reserved.

Summarize with AI



© Cord Technologies, Inc.
© Cord Technologies Limited